

M3/M4 Metro Staff App

Domain model explained from the perspective of the metro staff using the app

1. The staff member opens the app

A staff member pulls out their phone and opens the app. The first thing they see is the metro network.

- **M3M4**: the whole metro system and the top of everything. It holds all the lines and all the scenarios.
- **MetroLine**: a single line, like M3 or M4.
- **LineStop**: a station's place on a line, like "stop number 5 on M3". This is how the app shows the stations in the right order.
- **Station**: an actual station. Almost everything in the app happens *at* a station, so many classes point to it.
- **TrackSegment**: a piece of track between two stations (A → B). It lets the app pinpoint where on the line a problem is.

2. Something goes wrong, so they go to Scenarios

*Say there's a signal failure between two stations. The staff member taps **Scenarios** and finds the one that matches.*

- **Scenario**: a ready-made plan for a type of incident, like "track blocked between X and Y". It covers specific track segments, so it's clear which part of the network it's about.
- **ScenarioVersion**: scenarios get updated over time, so each one has versions. The app always shows the **current version**, and older ones are kept as history. The version holds the actual content of the plan.

3. They open the scenario and see how the trains are running

The first thing in the plan is how train traffic runs during the incident.

- **OperatingPlan**: the part of the plan about how the trains run.
- **OperatingPattern**: one way the trains run, either back and forth (**PENDULUM**) or in a loop (**ROUNDTRIP**), with route code, track, destination, a description, and the minimum and maximum number of trains.
- **PatternStop**: the stations that pattern stops at, in order.

This gives the staff member a quick overview of which parts of the line still have service and how the trains are running.

4. They see what has to happen at each station

Next, the plan shows what needs doing at each affected station.

- **StationAssignment**: the job at one specific station, with notes. This answers “what do I do at my station?”
- **StaffingRequirement**: how many people are needed there (minimum and maximum).
- **Action**: a step in the station checklist, with its order, the instruction, and a trigger condition (when to do it). For example: “1. When trains stop, close the platform and guide passengers to the exit.”
- **GeneralInstruction**: numbered instructions that apply to everyone, no matter which station they’re at.

This means staff don’t have to remember procedures off the top of their head. The checklist is right there.

5. They see what to tell passengers

- **PassengerInformation**: the message to give passengers, in which language, and on which channel: **PA** (the loudspeakers) or **PID** (the information screens).

Staff get the exact wording, so passengers hear the same thing everywhere.

6. The scenario is activated for real

When the incident is actually happening, the scenario is started.

- **ScenarioActivation**: the “live” use of a scenario. It records a description of the incident, when it started and ended, and whether it’s **ACTIVE** or **ENDED**. It’s tied to the exact version that was used, so afterwards you can see which plan was followed.

7. Staff are sent out and tracked

Staff members get sent to stations, and their phone shows where they need to go.

- **StaffDeployment**: one staff member being sent to one station assignment. Its status goes **AWAITING_RESPONSE** → **EN_ROUTE** → **AT_STATION**, with a timestamp for each step (assigned, responded, arrived).
- **StaffMember**: the person being sent (staff ID and name).

The staff member knows where to go and what to do when they get there. The control room can follow who has responded, who is on the way, and who has arrived.

The grey boxes (fixed choices)

These are enums, meaning fixed lists of allowed values:

Enum	Allowed values
ActivationStatus	ACTIVE / ENDED
DeploymentStatus	AWAITING_RESPONSE / EN_ROUTE / AT_STATION
OperationType	PENDULUM / ROUNDTRIP
PassengerInfoChannel	PA / PID